Xiaohua Cai

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Software Engineer with experience building full-stack web applications, REST APIs, authenticated workflows, and data-driven interfaces using React, Next.js, TypeScript, Node.js, PHP/Laravel, and relational databases. M.S. in Computer Science from Georgia Tech with a 4.0/4.0 GPA.

EDUCATION

Master of Science in Computer Science, Georgia Institute of Technology, GPA 4.0/4.0 2026
Master of Science in Biotechnology, University of Alberta, GPA 3.7/4.0 2018

SKILLS

Languages: Python, Java, JavaScript, PHP, TypeScript, SQL, HTML/CSS/SASS
Frameworks & Libraries: React, Angular, Next.js, Node.js, Express, Laravel, Socket.IO, Material UI, Tailwind CSS, Bootstrap
Databases & APIs: PostgreSQL, MySQL, MongoDB, Prisma, GraphQL, REST
Infrastructure & Tools: AWS, Docker, Docker Compose, Nginx, Git/GitHub, GitHub Actions, CI/CD, Postman, Ubuntu Linux

WORK EXPERIENCE

Software Development Engineer December 2022 - June 2024
Affinity Bridge Inc.

- Developed React frontend features and PHP REST APIs for the NovaSafe application, supporting thousands of users; reduced SQL execution time by approximately 20% through query restructuring, join and filter optimization, and targeted indexing; improved large-dataset performance with pagination.
- Designed REST API endpoints, middleware, relational data models, and caching strategies that reduced redundant data fetching and improved application performance.
- Built the Environmental Conventions Index application with Next.js, developing interactive geographic maps with React Simple Maps and responsive scatter and line charts with ECharts. Built reusable React UI components and custom hooks, integrated frontend workflows with REST APIs, and centralized shared data-fetching and state logic to reduce duplication and improve maintainability across the application.
- Developed 30+ automated tests with Playwright and React Testing Library for important application features and UI components, reducing manual testing time by approximately 25% before each release.
- Supported production deployments and application releases by investigating build, configuration, API, and runtime failures and implementing fixes to improve application stability and release reliability.
- Built an internal authenticated architecture-review application using Next.js, TypeScript, Prisma, and NextAuth, replacing Word-based reviews with a centralized workflow for 20+ users.

Software Development Engineer January 2022 - December 2022
AMI RockChain Inc.

- Designed and implemented PHP Laravel REST APIs, PostgreSQL relational schemas, migrations, and CRUD workflows using MVC architecture; built authentication and role-based authorization to enforce access permissions across different user types and protected application workflows.
- Developed reusable, responsive UI components and user workflows with AngularJS, TypeScript, and Nebular UI, building maintainable production features across the application; Built interactive mapping and data-visualization features using Leaflet, Google Maps API, and ECharts, and integrated the Mailchimp API to support subscriber-management workflows.
- Deployed web applications to AWS, managed production releases, and diagnosed and resolved deployment issues.

Software Development Engineer (Contract) September 2021 - February 2022
Explorer Hop

- Built a full-stack internal web application using React, TypeScript, Material UI, Node.js, Express, and PostgreSQL, implementing REST APIs, backend business logic, and relational data workflows.
- Developed Shopify admin app features using React, Shopify App Bridge, and Polaris; integrated the Shopify GraphQL API and implemented webhooks for product-order events.

Research and Development Scientist January 2019 - August 2021
SBI BioEnergy Inc.

- Developed VBA automation tools to calculate mass balances for 10+ lab-scale reactors, reducing repetitive manual calculations and supporting experimental analysis; Analyzed, processed, and visualized experimental data using Python.

PROJECTS

Distributed Real-Time Chat Application 2026
Tech stack: React, Node.js, Express, Socket.IO, Redis, PostgreSQL, Nginx, Docker

- Re-architected a previously built single-server real-time chat application into a horizontally scalable distributed system, running multiple Socket.IO backend instances behind an Nginx load balancer.
- Implemented Redis Pub/Sub for cross-instance event propagation and Redis-backed shared room/presence state; persisted message history in PostgreSQL and containerized the distributed stack with Docker Compose.
- Built a concurrent WebSocket load-testing script and validated 200 simultaneous clients and 40,000 message deliveries with zero errors; verified continued availability for new connections after terminating one backend instance.